Liang-Ching Tao

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Research Interests

Learning-Guided Planning and Search; Sequential Decision-Making; Reinforcement Learning; Multi-Agent Planning and Coordination; Representation Learning and World Models.

Education & Service

National Chung Hsing University (NCHU), Taiwan — M.S. in Computer Science and Engineering 2026

Advisor: Prof. Pi-Chung Wang. Graduate GPA: 3.75/4.00. Thesis: *A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.*

Mandatory Military Service — Private, Artillery Battalion, 249th Brigade, R.O.C. Army (Taiwan) 2024

National Chengchi University (NCCU), Taiwan — B.S. in Computer Science 2023

Research Experience

Learning-Guided Search for Multi-Agent Planning Sep. 2026–present

- Extending LP-BTS to coordinated multi-agent planning, using multi-charger scheduling as a structured testbed for graph-based policy representations, canonical actions, action-space reduction, and neural-guided search over the enlarged joint action space.

Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging (LP-BTS) 2026–present

- Formulated one-to-many mobile charging as planning over a state-dependent action space of physical charging stops that is rebuilt from the live geometry at every decision and exceeds 10^3 actions.
- Built LP-BTS: a graph proposal policy samples up to 32 candidate actions for search, a learned value critic evaluates leaves, and edge-budgeted PUCT compares simulated action sequences before an action is committed.
- Evaluated LP-BTS on a prespecified, sealed set of 30 scenarios; it improved survival by 12.29 percentage points over a trained, source-derived deep RL reconstruction under the common-physics protocol. Preprint [P1].

HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling — M.S. thesis, NCHU 2025–2026

- Formulated coupled charging, routing, and allocation as a structured sequential decision problem; developed a two-level architecture in which Q-learning selects regions or scheduling decisions and force-aware routing determines charger movement. Preprint [P2]; related conference presentations [C2] and [C5].

Preprints

[P1] **Liang-Ching Tao**, Pi-Chung Wang. "Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging." arXiv:2609.17429 (cs.LG; cross-list cs.AI), Sep. 2026.

[P2] **Liang-Ching Tao**, Pi-Chung Wang. "HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks." arXiv:2609.13901 (cs.NI; cross-list cs.LG), Sep. 2026.

Conference Papers and Presentations

MAMM 2026 presentation status and awards are noted below; one-page abstracts are linked where available, and full texts are available on request.

[C1] **Liang-Ching Tao**, Wei-Cheng Tai, Pi-Chung Wang. "Adaptive Charging Threshold and Detour Scheduling for Wireless Rechargeable Sensor Networks via Deep Reinforcement Learning." *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Oral presentation. [abstract]

[C2] **Liang-Ching Tao**, Pi-Chung Wang. "HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks." *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Poster presentation. **Best Poster Presentation.** Extended English version of [C5]; preprint [P2]. [arXiv]

[C3] **Liang-Ching Tao**, Po-Cheng Fu, Pi-Chung Wang. "DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks." *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Accepted for poster presentation. [abstract]

[C4] **Liang-Ching Tao**, Wei-Chen Lin, Pi-Chung Wang. "CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks." *MAMM 2026*, Sun Moon Lake, Taiwan, Nov. 5–7, 2026. Poster presentation. **Excellent Poster Presentation.** [abstract]

[C5] **Liang-Ching Tao**, Pi-Chung Wang. "A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks" (in Chinese). *2026 MC&WASN Conference*, National Chung Hsing University, Taiwan, Aug. 24–25, 2026. Presented as a poster. [PDF]

Other Research Experience

Research Assistant — Raman Spectral Analysis and Detection System, NCHU 2025–2026

- Built the full pipeline: spectral preprocessing, a 1D residual-convolutional encoder with multi-head attention, and binary Sudan Red I–IV detectors served through a FastAPI/Next.js application. [code]

Research Assistant — Citrus Crop Disease and Pest Recognition, NCHU 2025–2026

- Trained and compared EfficientNetV2-S and ResNet50 for class-imbalanced citrus disease and pest recognition; applied Focal Loss, MixUp/CutMix, and Grad-CAM; exported models to ONNX for inference. [code]

Teaching Experience

Teaching Assistant — Computer Networks (Undergraduate), NCHU 2025
Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — Computer Networks (Professional Master's Program), NCHU 2025
Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — New Trends in Network Technology (Graduate Course), NCHU 2025
Instructor: Prof. Pi-Chung Wang. Assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Technical Skills & Graduate Coursework

Machine Learning & AI: Reinforcement Learning; Deep Learning; Graph-based Learning; Planning & Search (MCTS/PUCT).
Programming & Frameworks: Python; C/C++; PyTorch; NumPy; pandas; scikit-learn; FastAPI; Flask; Next.js; JavaScript/TypeScript.
Systems & Tools: Linux; Git/GitHub; CUDA; Docker; ONNX; Kafka; LaTeX.
Graduate Coursework — AI & Data: Deep Reinforcement Learning (A−); Advanced Artificial Intelligence (A−); Natural Language Processing (A); Internet of Things (IoT) Data Analysis and Applications (A+).
Graduate Coursework — Systems & Hardware: Advanced VLSI Design (A+); Digital Testing (A); System-on-chip Testing (A−); New Trends in Network Technology (A−).